

Module 1 Lec 5

Interconnection of components

Interconnection of components

- The functional components of a computer need to communicate with each other to perform a task.
- Therefore, computer is a network of components.
- The collection of paths connecting the various components is called the *interconnection structure*. The design of this structure will depend on the exchanges that must be made among modules.

Input and output of a Memory Component

- Two Operations

- Read

- Input

- Enable Read Control signal
 - Provide the address on address bus from where the data to read

- Output

- Data from the given address

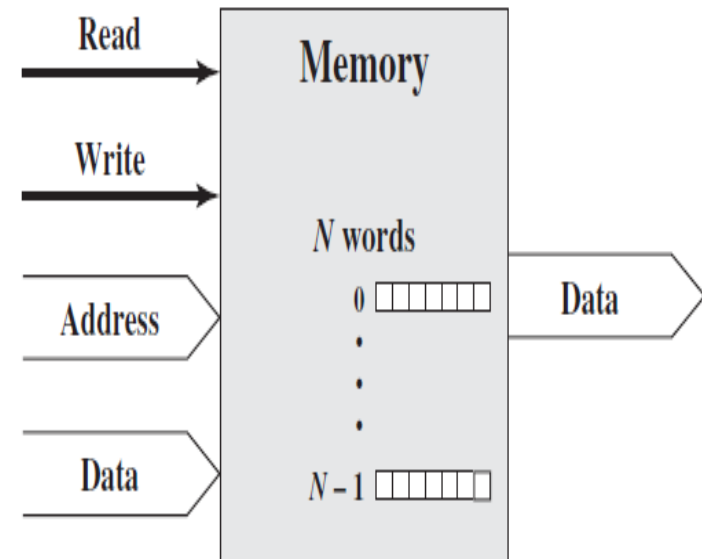
- Write

- Input

- Enable Write control signal
 - Provide the address on address bus where the data to be written
 - Provide the data to be written on data bus

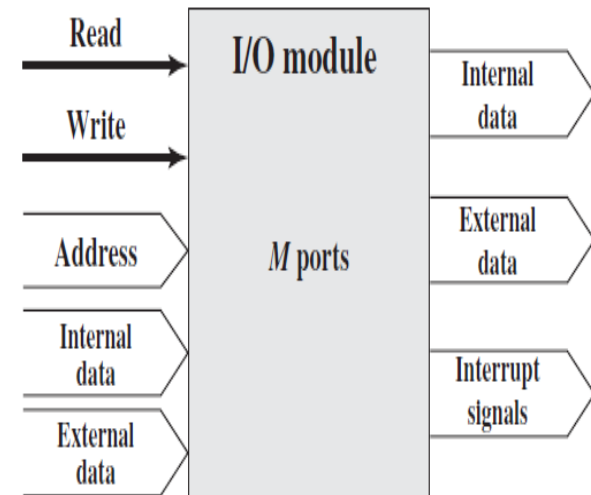
- Output

- -



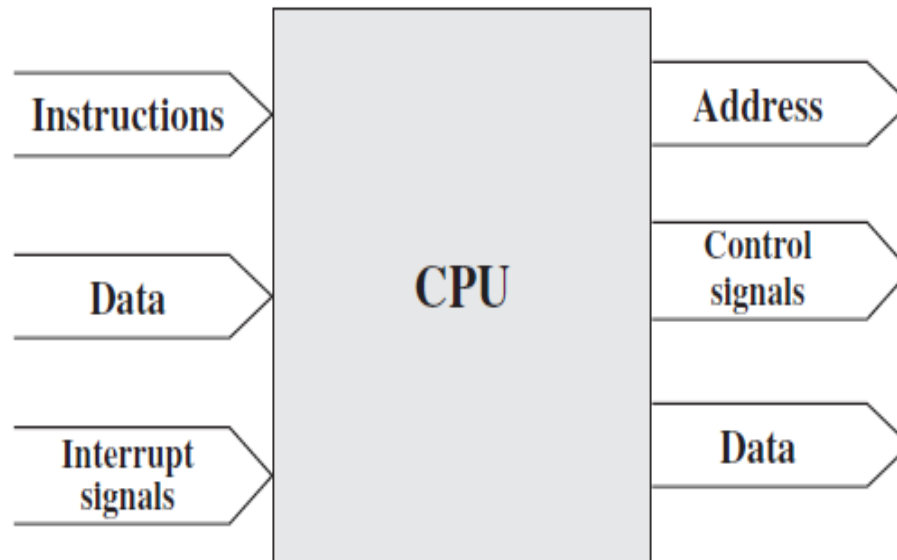
Input and Output of I/O Module

- I/O Module may control more than one external device.
- Each of the interface to an external device is referred as a *port* and given each a unique address (e.g., 0, 1, . . . ,M− 1)
- Finally, an I/O module may be able to send interrupt signals to the processor.



Input and Output of Processor component

- The processor reads in instructions and data, writes out data after processing, and uses control signals to control the overall operation of the system.
- It also receives interrupt signals.



Interconnections

- The interconnection structure must support the following types of transfers:
 - Memory $\leftrightarrow$ Processor
 - I/O $\leftrightarrow$ Processor
 - I/O $\leftrightarrow$ Memory (DMA)

Bus Interconnection

- It is a path way connecting two or more devices.
- A bus consists of multiple communication pathways, or lines.
- Each line is capable of transmitting signals representing binary 1 and binary 0.
- Transmitting a word across a single line (bit by bit serially)
- Several lines of a bus can be used to transmit binary digits simultaneously (in parallel).
 - For example, an 8-bit unit of data can be transmitted over eight bus lines.
- A bus that connects major computer components (processor, memory, I/O) is called a *system bus*



Bus

- The lines can be classified into three functional groups: **data, address, and control lines**
- **Data Lines – Data bus** – provide a path for moving data among system modules.
 - The data bus may consist of 32, 64, 128, or even more separate lines, the number of lines being referred to as the *width* of the data bus.
 - If the data bus is 32 bits wide and each instruction is 64 bits long, then the processor must access the memory module twice during each instruction cycle.

Address Bus

- **Address Lines – Address Bus** – Used to designate the source or destination of the data on the data bus.
- For example, if the processor wishes to read a word (8, 16, or 32 bits) of data from memory, it puts the address of the desired word on the address lines.
- Typically, the higher-order bits are used to select a particular module on the bus, and the lower-order bits select a memory location or I/O port within the module.
- For example, on an 8-bit address bus, address 01111111 and below might reference locations in a memory module with 128 words of memory, and address 10000000 and above refer to devices attached to an I/O module.

Control Bus

- The **control lines** are used to control the access to and the use of the data and address lines.
- Because the data and address lines are shared by all components, there must be a means of controlling their use.
- Control signals transmit both command and timing information among system modules.
- Timing signals indicate the validity of data and address information.
- Command signals specify operations to be performed.

Typical control lines include

- **Memory write:** Causes data on the bus to be written into the addressed location
- **Memory read:** Causes data from the addressed location to be placed on the bus
- **I/O write:** Causes data on the bus to be output to the addressed I/O port
- **I/O read:** Causes data from the addressed I/O port to be placed on the bus
- **Transfer ACK:** Indicates that data have been accepted from or placed on the bus
- **Bus request:** Indicates that a module needs to gain control of the bus
- **Bus grant:** Indicates that a requesting module has been granted control of the bus
- **Interrupt request:** Indicates that an interrupt is pending
- **Interrupt ACK:** Acknowledges that the pending interrupt has been recognized
- **Clock:** Is used to synchronize operations
- **Reset:** Initializes all modules

The operation of the bus

- If one module wishes to send data to another, it must do two things:
 - Obtain the use of the bus, and
 - Transfer data via the bus.
- If one module wishes to request data from another module, it must
 - Obtain the use of the bus, and
 - Transfer a request to the other module over the appropriate control and address lines.
 - It must then wait for that second module to send the data.

I/O Bus and Memory Bus

Functions of Buses

MEMORY BUS is for information transfers between CPU and the Main Memory

I/O BUS is for information transfers between CPU and I/O devices through their I/O interface

Many computers use a common single bus system for both memory and I/O interface units

- separate control lines for each function

- common control lines for both functions

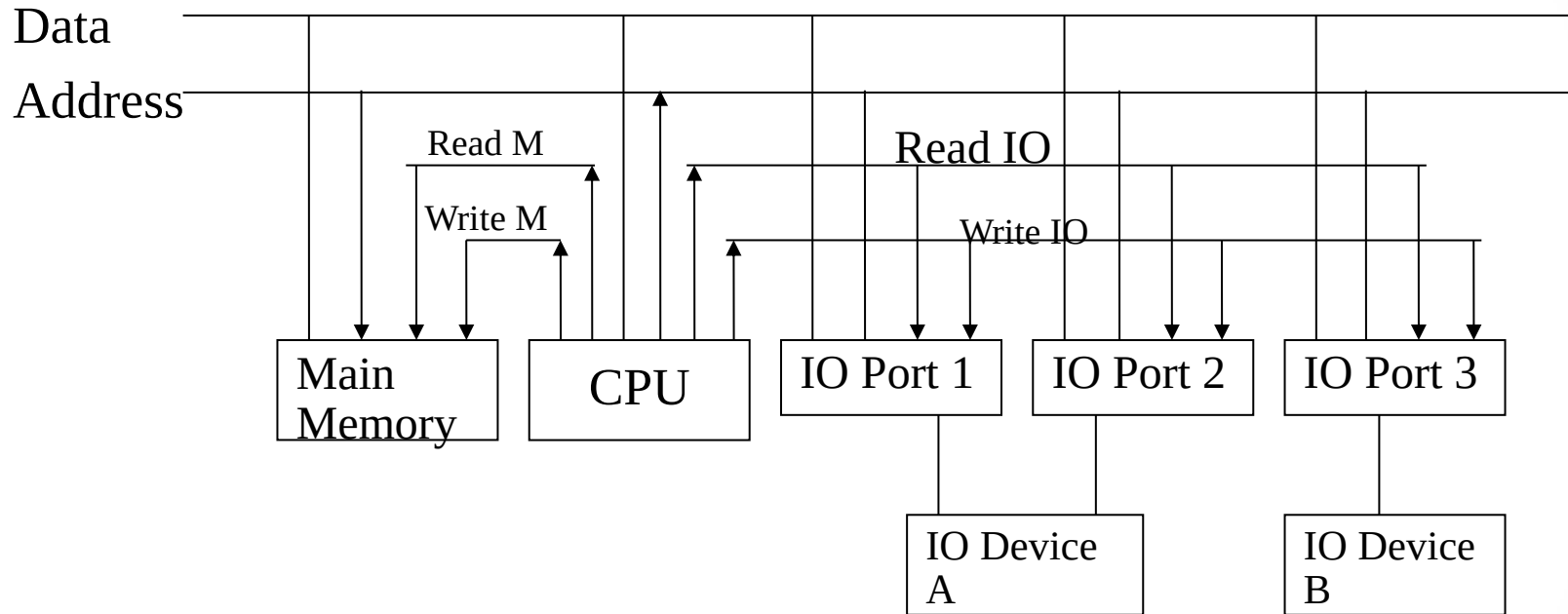
Some computer systems use two separate buses,

- one to communicate with memory and the other with I/O interfaces

Isolated I/O – I/O Mapped I/O

- Separate I/O read/write control lines in addition to memory read/write control lines
- Separate (isolated) memory and I/O address spaces
- Distinct input and output instructions
- When CPU fetches and decodes the opcode of an I/O instruction, it places the address into the common address lines.
- Also enables read/write control lines => the address in the address lines is for interface register and not for a memory word.

Isolated I/O



Memory Mapped I/O

- A single set of read/write control lines
- No distinction between memory and I/O transfer
- Memory and I/O addresses share the common address space
 - reduces memory address range available
 - Memory addresses `xffff0000` and above are used for I/O devices
- No specific input or output instruction
- The same memory reference instructions can be used for I/O transfers
- when the bus sees certain addresses, it knows they are not memory addresses, but are addresses for accessing I/O devices.

References

- Computer Architecture by William stallings